

Jiapeng Zhang 张嘉鹏

Ph.D. Student · Department of Computer Science and Technology, Tsinghua University

✉ jiapengzhang@mails.tsinghua.edu.cn

🌐 zjp-shadow.github.io

🔗 [zjp-shadow](#)

👤 [Google Scholar](#)

EDUCATION

Tsinghua University, Beijing, China

Sep. 2024 – Present

Ph.D. Student, Department of Computer Science and Technology

- **GPA:** 3.95 · **Rank:** 27/158
- **Advisor:** Prof. Shi-Min Hu (Institute of Human-Computer Interaction and Media Integration)
- **Research Focus:** 3D Generation, Animation, Graphics Simulation & Embodied AI, and AI Infra.

Tsinghua University, Beijing, China

Sep. 2020 – Jul. 2024

B.S. in Information and Computing Science, Zhili College

- **Rank:** 21/30

SELECTED PUBLICATIONS

[1] **SkinTokens: A Learned Compact Representation for Unified Autoregressive Rigging**

Jiapeng Zhang, Chengfeng Pu, Menghao Guo, Yan-Pei Cao, Shi-Min Hu

ACM Transactions on Graphics (TOG), 2026 **Accepted with Minor Revisions**

[Project](#)

[arXiv](#)

[Code](#)

Introduced SkinTokens, a learned discrete representation of skinning weights. By framing skinning as a sequence prediction task, the unified TokenRig framework jointly models skeletons and skin deformations, achieving substantial improvements on rigorous out-of-distribution geometry evaluations.

[2] **One Model to Rig Them All: Diverse Skeleton Rigging with UniRig**

Jiapeng Zhang, Chengfeng Pu, Menghao Guo, Yan-Pei Cao, Shi-Min Hu

ACM Transactions on Graphics (TOG), SIGGRAPH 2025

[Project](#)

[arXiv](#)

[Code](#)

Proposed UniRig, a unified framework for automatic skeletal rigging built on a large autoregressive model and a novel Skeleton Tree Tokenization. Trained on Rig-XL, a new large-scale dataset, it efficiently generates topologically valid skeletons and far outperforms existing state-of-the-art tools across diverse models.

[3] **CharacterGen: Efficient 3D Character Generation from Single Images with Multi-View Pose Canonicalization**

Haoyang Peng, **Jiapeng Zhang**, Menghao Guo, Yan-Pei Cao, Shi-Min Hu

ACM Transactions on Graphics (TOG), SIGGRAPH 2024, 43(4): 1–13

[Project](#)

[arXiv](#)

[Code](#)

Presented an efficient framework that lifts a single image into a pose-canonicalized 3D character mesh. It combines an image-conditioned multi-view diffusion model for pose calibration with a transformer-based sparse-view reconstruction network, producing geometrically and texturally robust results ready for animation.

[4] **Transferring the Intelligence of VLMs to Robotic Control**

Menghao Guo, Zhehan Mo, Jiajun Wang, Yi Zhang, Kejin Wang, Yixuan Deng, **Jiapeng Zhang**, Yongming Rao, Shi-Min Hu

arXiv preprint arXiv:2609.22966, 2026

[Project](#)

[arXiv](#)

Introduced RoboDawn, a human-intuitive interface that lets an agentic VLM control robots in a closed loop through discrete translation, rotation and gripper commands, grounded by a few in-context demonstrations. Without task-specific robot training it reaches state-of-the-art success rates on RoboTwin 2.0 C2R and RoboDojo, and transfers to a real Franka arm.

OPEN SOURCE PROJECTS

UniRig & SkinTokens

[UniRig](#)

[SkinTokens](#)

2,200+ Stars

Project Maintainer & Core Contributor

- Maintain the official open-source implementations of UniRig (SIGGRAPH 2025, 1.8k+ stars) and SkinTokens (TOG 2026, 450+ stars), releasing pre-trained skeleton and skinning models that rig meshes of every genre, from humanoids to fictional creatures and inanimate objects.
- Released the curated Rig-XL dataset and the unified TokenRig pipeline that predicts skeletons and skinning weights in a single autoregressive model, scaling automatic rigging for the 3D community.

CharacterGen

GitHub

830+ Stars

Project Maintainer & Core Contributor

- Designed and maintained the rendering and inference tools behind single-image 3D character lifting, which resolves pose ambiguity through canonical A-pose generation.
- Open-sourced multi-view diffusion synthesis and texture back-projection, widely used by 3D anime avatar creators.

Tsinghua Jittor Framework

Jittor

JittorInfer

3,200+ Stars

Core Member & Head of the JittorInfer Framework

- Engineered JittorInfer, a high-performance C++ LLM inference framework adapted to domestic AI hardware such as Huawei Ascend NPUs.
- Led system-level optimizations including multi-stage operator fusion and end-to-end resource scheduling, achieving large throughput gains over Ascend-vLLM when serving frontier models such as DeepSeek.

HONORS AND AWARDS

Rising Star Pioneer Award, Huawei Kunpeng & Ascend Developer Conference (KADC)	2026
Comprehensive Scholarship for Ph.D. Students, Tsinghua University	2025
Silver Medal, National Olympiad in Informatics (NOI)	2019
Gold Medal, China Team Selection for Informatics (CTS)	2019

INTERNSHIP EXPERIENCE

Tencent Hunyuan

2026

Research Intern, Video Generation · Tsinghua Summer Social Practice

- Studied how the metrics of video VAEs relate to the generation performance of flow-based video models.

Tencent

2024

Research Intern

- Conducted research on 3D multi-modal understanding.

Huawei

2023

Software Engineering Intern

- Developed system-level optimizations for sparse matrix multiplication targeting the SME (Scalable Matrix Extension) instruction set on Kunpeng chips.